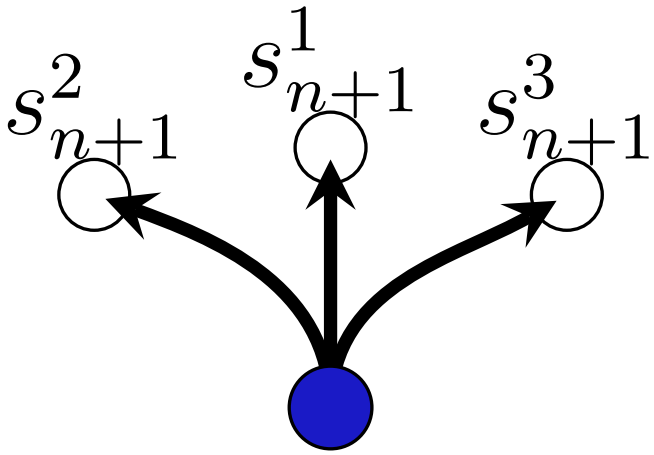
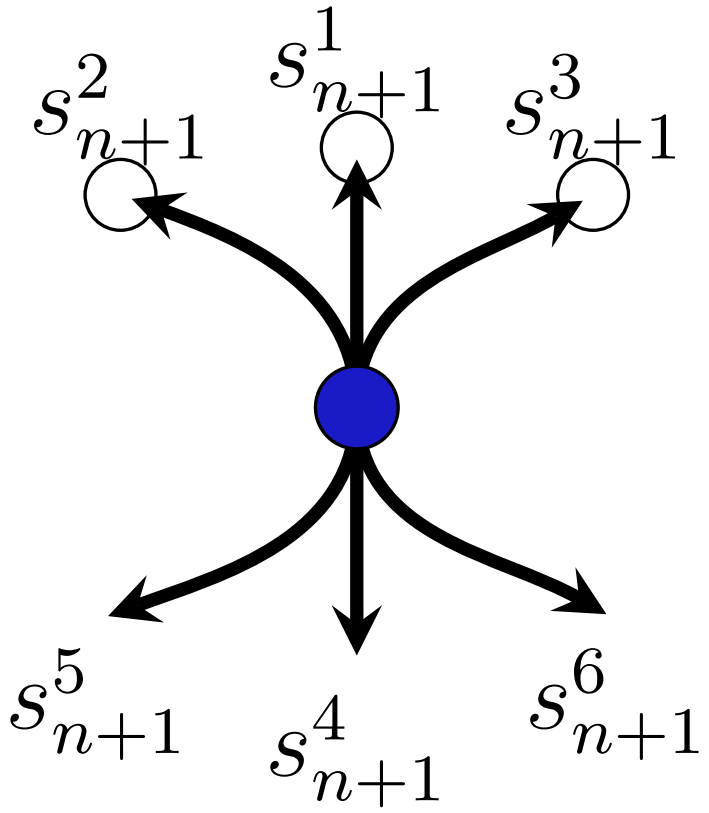
$$S_n = (x_n, \theta_n)$$